Tanush Chawla

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LinkedIn | GitHub | LeetCode

AI/ML and Data Analytics Enthusiast with strong foundations in machine learning, statistical analysis, and data-driven problem solving. Experienced in exploratory data analysis (EDA), data preprocessing, feature engineering, and building machine learning models using Python, SQL, and analytical tools. Skilled in extracting insights from structured datasets and developing dashboards to support data-driven decision making.

EDUCATION

VIT Bhopal University, Madhya Pradesh, India **CGPA: 8.59**
B.Tech in Computer Science Engineering (AI & ML)
Relevant Coursework: Applied Machine Learning, Artificial Neural Networks, Deep Learning, Reinforcement Learning, DBMS, Data Structures, Data Mining

Senior Secondary (Class XII) **Percentage: 90%**
CBSE Board — Blue Bells Model School, Gurugram, Haryana

Secondary (Class X) **Percentage: 90%**
CBSE Board — Our Lady of Fatima Convent Secondary School, Gurugram, Haryana

TECHNICAL SKILLS

Programming: Python, SQL, C++, JavaScript

Machine Learning: Regression, Classification, Clustering (K-Means), Feature Engineering, Model Training

Data Analysis: Pandas, NumPy, EDA, Statistical Analysis, Data Cleaning

Visualization: Power BI, Excel (Pivot Tables, Power Query, XLOOKUP)

Databases: MySQL, PostgreSQL, MongoDB

Tools: Git, REST APIs, Postman

Web: React.js, Node.js, Express.js

PROJECTS

GGN Home – AI Powered Real Estate Platform
Python, SQL, React.js, Node.js, Express.js, MongoDB, Power BI
- Built a **full-stack real estate platform** supporting property listings, analytics workflows, and intelligent search.
- Implemented **role-based dashboards** for Admin, Agents, and Users enabling property management and listing control.
- Developed an **AI-based property search and recommendation system** matching listings based on user preferences.
- Performed **exploratory data analysis (EDA)** and applied regression and clustering techniques to analyze real estate pricing trends.
- Built backend services using **Node.js and Express.js** with MongoDB for scalable data storage and querying.

Network Intrusion Detection System (NIDS)
Python, Scikit-learn, Flask, Docker, NSL-KDD Dataset
- Built an end-to-end **Network Intrusion Detection System** using machine learning to classify network traffic into normal and multiple attack categories (DoS, Probe, R2L, U2R), achieving high detection accuracy.
- Implemented **data preprocessing, feature engineering, and model training** on the NSL-KDD dataset for robust multi-class classification.
- Deployed the solution as a **Flask-based web application** with **Docker support** for scalable and portable inference.

RAG-Based Intelligence Research Agent
Python, LangChain, Vector Databases, Large Language Models
- Developed an **AI-powered RAG-based Research Agent** integrating large language models with vector-based semantic retrieval to generate accurate, context-aware responses.
- Designed an end-to-end pipeline for **document ingestion, embedding generation, and retrieval** using LangChain and vector databases.
- Improved **factual grounding and reduced hallucinations** in LLM outputs through structured retrieval-augmented generation.

EXPERIENCE

Software Development Intern (SDE) April 2025 – May 2025
Bluestock Fintech
- Contributed to backend development for fintech applications using **Node.js and REST APIs**.
- Worked with **MongoDB** to design and query financial datasets supporting application workflows.

IT Intern June 2025 – August 2025
Konverge Technologies Pvt. Ltd., Gurugram
- Analyzed IT operations datasets to identify incident patterns and service trends.
- Performed **data cleaning, preprocessing, and exploratory analysis** using Python and Excel.
- Applied basic **machine learning techniques** for ticket categorization and priority prediction.

ACHIEVEMENTS

- Solved **500+ Data Structures and Algorithms problems** on LeetCode and HackerRank, strengthening algorithmic problem solving.

CERTIFICATIONS

- Applied Machine Learning with Python — University of Michigan (Coursera)
- Python Essentials — Python Institute
- Google IT Support Professional Certificate — Google (Coursera)